

FinWealth Advisors — Data Warehouse

Source to Target Mapping Document

Overview

This document describes the complete source to target mapping for the FinWealth Advisors Data Warehouse. It covers all 11 source CSV files, 11 staging tables, 6 dimension tables, and 5 fact tables. All transformations are implemented via Azure Data Factory (ADF) and SQL Stored Procedures.

Dimension Tables

Dim_Advisor

Source: *Advisors.csv* → *stg_advisors*

Source Column	Source Table	Target Column	Target Table	Transformation
AdvisorID	stg_advisors	advisor_id	Dim_Advisor	Direct copy — natural key
AdvisorName	stg_advisors	advisor_name	Dim_Advisor	ISNULL(AdvisorName, 'Unknown')
Region	stg_advisors	region	Dim_Advisor	ISNULL(Region, 'Unknown')
JoinDate	stg_advisors	join_date	Dim_Advisor	ISNULL(JoinDate, '1900-01-01')
ExperienceYears	stg_advisors	experience_years	Dim_Advisor	ISNULL(ExperienceYears, 0)
—	Derived	ExperienceBand	Dim_Advisor	CASE: >=15=Senior, >=8=Mid-Level, >=3=Junior, else Trainee
—	System	advisor_sk	Dim_Advisor	IDENTITY(1,1) — surrogate key
—	System	is_current	Dim_Advisor	Default 1 — SCD Type 1
—	System	dwh_created_date	Dim_Advisor	GETDATE()
—	System	dwh_updated_date	Dim_Advisor	GETDATE() — updated on change

Dim_Client

Source: *Clients.csv* → *stg_clients*

Source Column	Source Table	Target Column	Target Table	Transformation
ClientID	stg_clients	client_id	Dim_Client	Direct copy — natural key
ClientName	stg_clients	client_name	Dim_Client	ISNULL(ClientName, 'Unknown')
Segment	stg_clients	segment	Dim_Client	ISNULL(Segment, 'Unknown')
RiskProfile	stg_clients	risk_profile	Dim_Client	ISNULL(RiskProfile, 'Unknown')
Country	stg_clients	country	Dim_Client	ISNULL(Country, 'Unknown')
ClientSince	stg_clients	client_since	Dim_Client	ISNULL(ClientSince, '1900-01-01')

Source Column	Source Table	Target Column	Target Table	Transformation
PrimaryAdvisorID	stg_clients	advisor_id	Dim_Client	ISNULL(PrimaryAdvisorID, 'Unknown')
—	Derived	ClientTier	Dim_Client	CASE Segment: Ultra HNI=Platinum, HNI/HNW=Gold, Mass Affluent=Silver, else Standard
—	Derived	TenureYears	Dim_Client	DATEDIFF(YEAR, ClientSince, GETDATE())
—	System	client_sk	Dim_Client	IDENTITY(1,1) — surrogate key

Dim_Account

Source: Accounts.csv → stg_accounts

Source Column	Source Table	Target Column	Target Table	Transformation
AccountID	stg_accounts	account_id	Dim_Account	Direct copy — natural key
ClientID	stg_accounts	client_id	Dim_Account	ISNULL(ClientID, 'Unknown')
AccountType	stg_accounts	account_type	Dim_Account	ISNULL(AccountType, 'Unknown')
OpenDate	stg_accounts	open_date	Dim_Account	ISNULL(OpenDate, '1900-01-01')
BaseCurrency	stg_accounts	base_currency	Dim_Account	ISNULL(BaseCurrency, 'Unknown')
Status	stg_accounts	status	Dim_Account	ISNULL(Status, 'Unknown')
—	Derived	CurrencyGroup	Dim_Account	CASE: USD/GBP/EUR/JPY=Major, INR/BRL/ZAR/MXN=Emerging, else Other
—	System	account sk	Dim Account	IDENTITY(1,1) — surrogate key

Dim_Security

Source: Securities.csv → stg_securities

Source Column	Source Table	Target Column	Target Table	Transformation
SecurityID	stg_securities	security_id	Dim_Security	Direct copy — natural key
Ticker	stg_securities	ticker	Dim_Security	ISNULL(Ticker, 'Unknown')
SecurityName	stg_securities	security_name	Dim_Security	ISNULL(SecurityName, 'Unknown')
AssetClass	stg_securities	asset_class	Dim_Security	ISNULL(AssetClass, 'Unknown')
Sector	stg_securities	sector	Dim_Security	ISNULL(Sector, 'Unknown')
Currency	stg_securities	currency	Dim_Security	ISNULL(Currency, 'Unknown')
Region	stg_securities	region	Dim_Security	ISNULL(Region, 'Unknown')

Source Column	Source Table	Target Column	Target Table	Transformation
—	Derived	AssetClassGroup	Dim_Security	CASE: Equity=Growth, Bond=Fixed Income, ETF=Passive, Mutual Fund=Managed, else Alternative
—	System	security_sk	Dim_Security	IDENTITY(1,1) — surrogate key

Dim_Benchmark

Source: *Benchmarks.csv* → *stg_benchmarks*

Source Column	Source Table	Target Column	Target Table	Transformation
BenchmarkCode	stg_benchmarks	benchmark_code	Dim_Benchmark	Direct copy — natural key
Description	stg_benchmarks	description	Dim_Benchmark	ISNULL(Description, 'Unknown')
Currency	stg_benchmarks	currency	Dim_Benchmark	ISNULL(Currency, 'Unknown')
—	System	benchmark_sk	Dim_Benchmark	IDENTITY(1,1) — surrogate key

dim_date

Source: *System Generated* — no source CSV

Source Column	Source Table	Target Column	Target Table	Transformation
—	System	date_sk	dim_date	FORMAT(dt, 'yyyyMMdd') — integer key
—	System	full_date	dim_date	Date spine 2015-01-01 to 2030-12-31
—	System	day_of_week	dim_date	DATENAME(WEEKDAY, dt)
—	System	month_name	dim_date	DATENAME(MONTH, dt)
—	System	quarter_name	dim_date	'Q' + CAST(DATEPART(QUARTER, dt) AS VARCHAR)
—	System	is_weekday	dim_date	CASE WHEN DATEPART(WEEKDAY,dt) IN (1,7) THEN 0 ELSE 1
—	System	is_month_end	dim_date	CASE WHEN dt = EOMONTH(dt) THEN 1 ELSE 0

Fact Tables

fact_holdings

Source: Holdings.csv + Prices.csv → stg_holdings + stg_prices

Source Column	Source Table	Target Column	Target Table	Transformation
AccountID	stg_holdings	account_sk	fact_holdings	Lookup Dim_Account.account_sk WHERE account_id = AccountID
SecurityID	stg_holdings	security_sk	fact_holdings	Lookup Dim_Security.security_sk WHERE security_id = SecurityID
ValuationDate	stg_holdings	valuation_date_sk	fact_holdings	Lookup dim_date.date_sk WHERE full_date = ValuationDate
Quantity	stg_holdings	quantity	fact_holdings	ISNULL(Quantity, 0)
ClosePrice	stg_prices	close_price	fact_holdings	JOIN stg_prices ON SecurityID + PriceDate
MarketValue	stg_holdings	market_value	fact_holdings	ISNULL(MarketValue, 0)
—	System	holding_sk	fact_holdings	IDENTITY(1,1) — surrogate key

Fact_Trades

Source: Trades.csv → stg_trades

Source Column	Source Table	Target Column	Target Table	Transformation
TradeID	stg_trades	trade_id	Fact_Trades	Direct copy — business key
AccountID	stg_trades	account_sk	Fact_Trades	Lookup Dim_Account.account_sk
SecurityID	stg_trades	security_sk	Fact_Trades	Lookup Dim_Security.security_sk
TradeDate	stg_trades	trade_date_sk	Fact_Trades	Lookup dim_date.date_sk
SettleDate	stg_trades	settle_date_sk	Fact_Trades	Lookup dim_date.date_sk (LEFT JOIN)
Side	stg_trades	side	Fact_Trades	ISNULL(Side, 'Unknown')
Price	stg_trades	price	Fact_Trades	ISNULL(Price, 0)
Quantity	stg_trades	quantity	Fact_Trades	ISNULL(Quantity, 0)
Amount	stg_trades	trade amount	Fact_Trades	ISNULL(Amount, 0)
—	Derived	gross_amount	Fact_Trades	ABS(ISNULL(Amount, 0))
—	Derived	TradeCategory	Fact_Trades	CASE Side: BUY=Purchase, SELL=Liquidation
—	Derived	TradeSize	Fact_Trades	CASE: >=100000=Large, >=10000=Medium, else Small

Fact_Cashflows

Source: *Cashflows.csv* → *stg_cashflows*

Source Column	Source Table	Target Column	Target Table	Transformation
CashflowID	stg_cashflows	cashflow_id	Fact_Cashflows	Direct copy — business key
AccountID	stg_cashflows	account_sk	Fact_Cashflows	Lookup Dim_Account.account_sk
CashflowDate	stg_cashflows	cashflow_date_sk	Fact_Cashflows	Lookup dim_date.date_sk
Type	stg_cashflows	cashflow_type	Fact_Cashflows	ISNULL(Type, 'Unknown')
Amount	stg_cashflows	amount	Fact_Cashflows	ISNULL(Amount, 0)
—	Derived	FlowDirection	Fact_Cashflows	CASE: Deposit/Interest/Dividend=Inflow, Withdrawal=Outflow, else Neutral

fact_fees

Source: *Fees.csv* → *stg_fees*

Source Column	Source Table	Target Column	Target Table	Transformation
FeeID	stg_fees	fee_id	fact_fees	Direct copy — business key
AccountID	stg_fees	account_sk	fact_fees	Lookup Dim_Account.account_sk
FeeDate	stg_fees	fee_date_sk	fact_fees	Lookup dim_date.date_sk
FeeType	stg_fees	fee_type	fact_fees	ISNULL(FeeType, 'Unknown')
Amount	stg_fees	fee_amount	fact_fees	ISNULL(Amount, 0)

Fact_Performance

Source: *Performance.csv* → *stg_performance*

Source Column	Source Table	Target Column	Target Table	Transformation
AccountID	stg_performance	account_sk	Fact_Performance	Lookup Dim_Account.account_sk
MonthStart	stg_performance	month_start_date_sk	Fact_Performance	Lookup dim_date.date_sk
BenchmarkCode	stg_performance	benchmark_sk	Fact_Performance	Lookup Dim_Benchmark.benchmark_sk (LEFT JOIN)
ReturnPct	stg_performance	return_pct	Fact_Performance	ISNULL(ReturnPct, 0)
BenchmarkReturnPct	stg_performance	benchmark_return_pct	Fact_Performance	ISNULL(BenchmarkReturnPct, 0)
ExcessReturnPct	stg_performance	excess return pct	Fact_Performance	ISNULL(ExcessReturnPct, 0)
CumulativeReturnPct	stg_performance	cumulative_return_pct	Fact_Performance	ISNULL(CumulativeReturnPct, 0)
—	Derived	PerformanceBand	Fact_Performance	CASE: >0.02=Strong Outperform, >0=Outperform, =0=In Line, <0=Underperform

NULL Handling Rules

Column Type	NULL Replacement	Example
Text / Name columns	ISNULL(col, 'Unknown')	Dim_Client.client_name, Dim_Advisor.region
Date columns	ISNULL(col, '1900-01-01')	Dim_Client.client_since, Dim_Account.open_date
Numeric / Amount cols	ISNULL(col, 0)	fact_holdings.quantity, Fact_Trades.price
Key / ID columns	WHERE col IS NOT NULL	All primary key columns — row rejected if NULL

SCD Strategy

All dimension tables use SCD Type 1 (overwrite). When source data changes, the existing row is updated in place. No history is kept. The `is_current` flag is always 1 for active records.